Problem A: Space Diving

An Aerodynamic and Thermodynamic Analysis of the Maximum Release Altitude for Space Diving

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Summary

The total mass of a space diver, pressure suit, and parachute does not by itself determine a survivable release altitude. We model vertical release near rocket apogee using a layered atmosphere, Mach-dependent drag, aerodynamic heating, a local flat-spin response, and finite-time parachute inflation followed by canopy descent. The three 2012 Red Bull Stratos peak speeds calibrate an effective frontal area. A digitized October speed–altitude curve checks trajectory shape, although that flight's peak also enters calibration. Nearby radiosonde records check the reference atmosphere only below about 30 km.

For an illustrative equipment design, we impose a recovery-temperature limit of 390 K, limits of 5 g_0 on free-fall aerodynamic proper acceleration and 5 g_0 on head radial acceleration, a modeled parachute-opening limit of 5 g_0, and a landing-speed limit of 6 m/s. The joint screening altitude is 56.36 km, governed by the thermal condition; peak downward speed there is 594.4 m/s. The free-fall load condition alone crosses at 105.58 km only after extending the atmospheric model beyond its nominal 86 km range. The assumed spin response crosses at 66.28 km.

These are model-design screens, not certified human or equipment tolerances. Suit material heating, true parachute opening shocks, and spin-control coefficients remain unmeasured for the proposed 190 kg assembly. Thus 56.36 km is a conditional result under the declared assumptions, not a universal safe maximum.

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1 Introduction

1.1 Background

The rapid development of crewed spaceflight, suborbital flight, and commercial space tourism has increased interest in emergency escape and life support in near space. Space diving has also evolved from an extreme sport into a platform for studying human physiology at altitude and supersonic aerodynamics.

In 2012, Felix Baumgartner completed two test space dives and a final mission as part of Red Bull Stratos. The mission summary report [1] documents altitude, speed, attitude, and parachute-opening records for all three flights, providing measurements with which to study descent from high altitude. Investigating the descent characteristics of a space-diving system with a specified mass therefore has clear engineering relevance.

1.2 Problem and Approach

We consider a rocket carrying a space diver vertically to a release altitude. The diver then exits wearing a pressure suit and carrying a parachute, with a total system mass of 190 kg, and descends to the surface. Our objective is to estimate the highest altitude from which this descent can be completed while preserving the diver's safety.

The aerodynamic study [2] uses a belly-down posture, with the body's longitudinal axis approximately horizontal, as the reference attitude for space diving. We assume that this posture is maintained throughout the descent and divide the process into three stages:

1. **Free fall.** Between release and main-canopy inflation, gravity and aerodynamic drag, which varies with altitude and speed, govern translation; the dynamics are developed in Section 4.2.1. At the same time, the deceleration of near-wall airflow raises the recovery temperature, as discussed in Section 4.2.2. Aerodynamic asymmetry can also induce attitude motion, which is considered in Section 4.3.

2. **Parachute inflation.** The effective canopy area grows over a finite interval, and the resulting drag rapidly decelerates the system. This stage is modeled in Section 4.4.1.

3. **Descent under the fully inflated canopy and landing.** Body and canopy drag act together as the speed approaches local equilibrium until the altitude reaches zero; Section 4.4.2 describes this stage.

Section 4 develops the physical models, Section 5 calibrates and examines them using publicly available Red Bull Stratos measurements, and Section 6 evaluates the hazards to determine the maximum admissible release altitude.

2 Assumptions

1. The atmosphere is stationary and varies only with altitude. Motion is approximately vertical, the initial vertical speed is zero, and total mass is constant. Horizontal drift, lift, Earth's rotation, and the rocket wake are neglected.

2. The ground is at sea level, so parachute-opening and landing altitudes share the same geometric reference.

3. The pressure suit remains sealed, with sufficient oxygen and pressure-maintenance capability for the entire computed descent. The parachute is intact and functional.

4. The space diver maintains the belly-down posture throughout the descent.

5. The inertia of the body and equipment is approximated by that of a uniform cylinder. The belly-down reference attitude is assumed to be stable in pitch, with small initial pitch disturbances and zero initial flat-spin angular velocity.

6. The space diver is well trained, can assess the descent state, opens the parachute promptly at the prescribed altitude, and reacts and adjusts posture within 1 s after detecting flat spin.

7. A space dive is considered successful if all loads experienced during descent remain within human tolerance and body temperature remains normal.

3 Notation

Altitude is positive upward, whereas descent speed is positive downward. Table 1 lists the principal quantities used in the model.

Table 1: Principal symbols, definitions, and units

Symbol	Definition	Unit
h	Geometric altitude	m
h_0	Release altitude	m
z	Geopotential altitude; entered in km in the atmospheric piecewise formulas	m
v	Downward descent speed	m/s
m	Total mass of the body, pressure suit, and parachute	kg
T	Atmospheric temperature	K
ρ	Air density	kg/m^3
C_D	Body drag coefficient	1

Table 1 (continued)

Symbol	Definition	Unit
A	Calibrated effective frontal area of the body and equipment	m^2
D	Aerodynamic drag on the body	N
T_{aw}	Adiabatic-wall recovery temperature	K
ω	Local flat-spin angular velocity	rad/s
I_z	Moment of inertia about the reference flat-spin axis	kg m^2
C_n	Flat-spin bias-moment coefficient	1
$C_{n\omega}$	Positive flat-spin damping coefficient	1
τ	Local elapsed time after the onset of the spin bias	s
B_c	Parachute drag area, $C_{D,c}A_c$	m^2
D_c	Parachute drag	N
τ_c	Inflation time	s
v_t	Local free-fall terminal speed	m/s
v_{land}	Descent speed immediately before ground contact	m/s
n_D	Free-fall aerodynamic specific force divided by g_0	1
n_o	Total aerodynamic specific force during opening divided by g_0	1
n_r	Radial acceleration in the local 1 s response divided by g_0	1

4 Physical Models

4.1 Atmosphere

We construct a layered atmosphere from the standard-atmosphere reference [3, Table 4]. Because its temperature layers are specified in geopotential altitude, geometric altitude must first be converted. Geopotential altitude represents the equivalent height under sea-level gravity for the same change in gravitational potential energy:

$$z = \frac{R_e h}{R_e + h}. \tag{4.1}$$

Here h is geometric altitude and $R_e = 6.357 \times 10^3$ km is the reference Earth radius. The corresponding gravitational acceleration is

$$g(h) = g_0 \left(\frac{R_e}{R_e + h} \right)^2. \tag{4.2}$$

With $g_0 = 9.807\,\mathrm{m/s^2}$, Eq. (4.1) ensures that the potential-energy increment satisfies $g(h)\,\mathrm{d}h = g_0\,\mathrm{d}z$.

The seven temperature gradients in Table 4 of the standard-atmosphere reference lead to

$$T[\mathrm{K}](z[\mathrm{km}]) = \begin{cases} 288.15 - 6.5z, & 0 \le z < 11, \\ 216.65, & 11 \le z < 20, \\ 216.65 + (z - 20), & 20 \le z < 32, \\ 228.65 + 2.8(z - 32), & 32 \le z < 47, \\ 270.65, & 47 \le z < 51, \\ 270.65 - 2.8(z - 51), & 51 \le z < 71, \\ 214.65 - 2(z - 71), & 71 \le z \le 84.852. \end{cases} \tag{4.3}$$

As shown in Figure 1, atmospheric temperature varies between layers; this structure also determines the altitude dependence of the speed of sound.

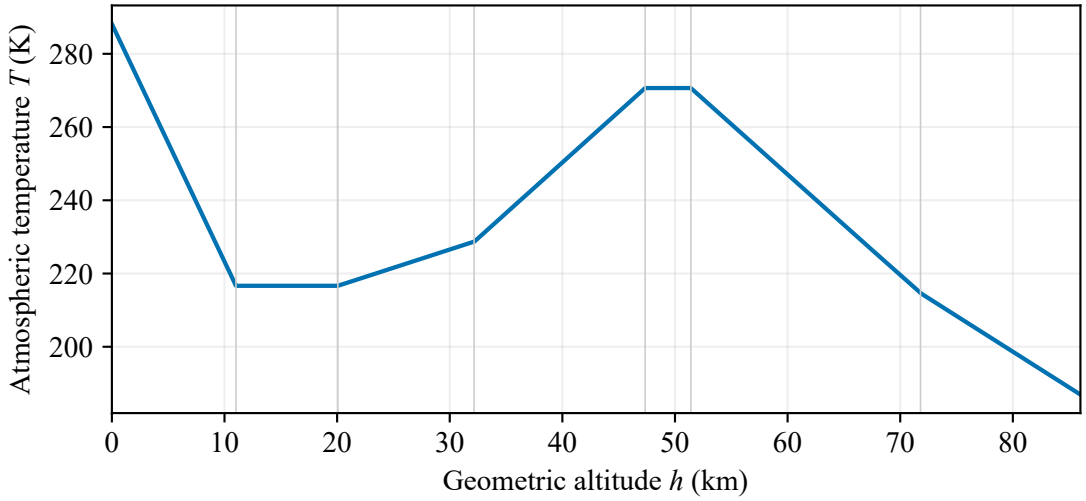


Figure 1: Temperature versus geometric altitude in the seven-layer standard atmosphere.

The specific gas constant for air is taken as $R = 287.1\,\mathrm{J/(kg\,K)}$. Hydrostatic equilibrium requires the pressure gradient to balance the weight of the air:

$$\frac{\mathrm{d}p}{\mathrm{d}h} = -\rho g(h). \tag{4.4}$$

The ideal-gas law is

$$p = \rho R T. \tag{4.5}$$

Combining Eqs. (4.1), (4.3), (4.4), and (4.5), starting from a sea-level pressure of 101.3 kPa and enforcing pressure continuity across layer interfaces, yields the density profile:

$$\rho[\mathrm{kg/m^3}](z[\mathrm{km}]) = \begin{cases} 1.225\left(1-\dfrac{6.5z}{288.15}\right)^{4.256}, & 0 \le z < 11, \\ 0.3639\, e^{-0.1577(z-11)}, & 11 \le z < 20, \\ 8.803\times10^{-2}\left(1+\dfrac{z-20}{216.65}\right)^{-35.16}, & 20 \le z < 32, \\ 1.323\times10^{-2}\left(1+\dfrac{2.8(z-32)}{228.65}\right)^{-13.20}, & 32 \le z < 47, \\ 1.428\times10^{-3}\, e^{-0.1262(z-47)}, & 47 \le z < 51, \\ 8.616\times10^{-4}\left(1-\dfrac{2.8(z-51)}{270.65}\right)^{11.20}, & 51 \le z < 71, \\ 6.421\times10^{-5}\left(1-\dfrac{2(z-71)}{214.65}\right)^{16.08}, & 71 \le z \le 84.852. \end{cases} \tag{4.6}$$

Converting back to geometric altitude through Eq. (4.1) produces the density profile in Figure 2. Density decreases by several orders of magnitude with increasing altitude, which is the main environmental reason for the contrast between high-altitude acceleration and low-altitude deceleration.

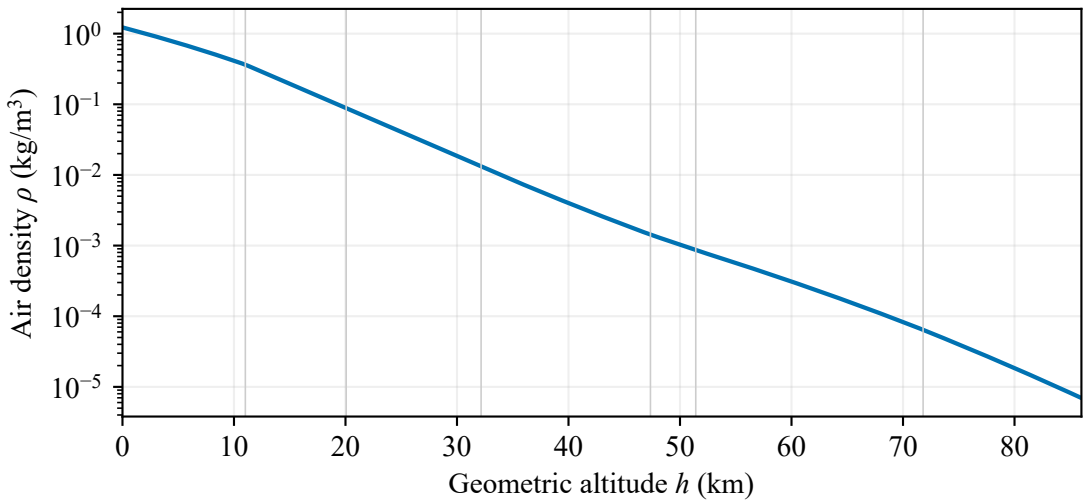


Figure 2: Standard-atmosphere density versus geometric altitude.

4.2 Free Fall

4.2.1 Forces and Vertical Motion

As illustrated in Figure 3, two external forces act vertically during free fall. We represent the system's translation by the motion of its center of mass.

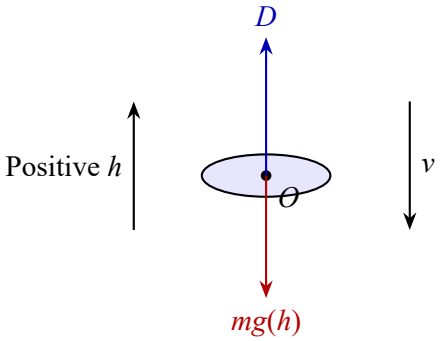


Figure 3: Forces at the center of mass and the direction of motion during free fall.

1. **Gravity.** The weight $mg(h)$ acts downward, with $g(h)$ defined by Eq. (4.2) and total mass $m = 190\,\text{kg}$.

2. **Aerodynamic drag.** The large characteristic dimensions of the body and canopy, together with the high descent speed, place the flow in an inertia-dominated continuum regime at high Reynolds number. Aerodynamic forces therefore scale with dynamic pressure, motivating the quadratic drag law

$$D(h) = \frac{1}{2}\rho(h)C_D(M)Av(h)^2. \tag{4.7}$$

Drag acts upward and represents the combined effects of pressure and skin-friction drag. The three aerodynamic parameters are specified as follows.

1. **Air density.** The density $\rho(h)$ follows Eq. (4.6) and the geopotential-altitude transformation, using the source described in Section 4.1.

2. **Drag coefficient.** As speed approaches the local speed of sound, compressibility becomes significant, and local shocks and changes in pressure distribution increase drag. Thus C_D depends on the ratio of speed to sound speed, namely the Mach number. The local speed of sound is

$$c(h) = \sqrt{\gamma RT(h)}.$$

The Mach number is defined as

$$M = \frac{v}{c(h)}.$$

Here $T(h)$ follows Eq. (4.3), and $\gamma = 1.4$ is the specific-heat ratio for air under the constant-specific-heat approximation. We adopt the drag relation inferred from measured flight data in the aerodynamic study [2]:

$$C_D(M) = \begin{cases} 0.600, & 0 \le M \le 0.6, \\ 0.600 + 0.550(M - 0.6)^2, & 0.6 < M \le 1.1, \\ 0.740 - 0.320(M - 1.1), & 1.1 < M \le 1.25, \\ 0.692, & M > 1.25. \end{cases} \tag{4.8}$$

The final branch is a constant extrapolation beyond the original data range. As shown in Figure 4, C_D rises and then falls across the transonic region.

3. **Frontal area.** The constant A is the effective aerodynamic area of the body and equipment. It is jointly calibrated against three peak-speed observations in Section 5.1.3.

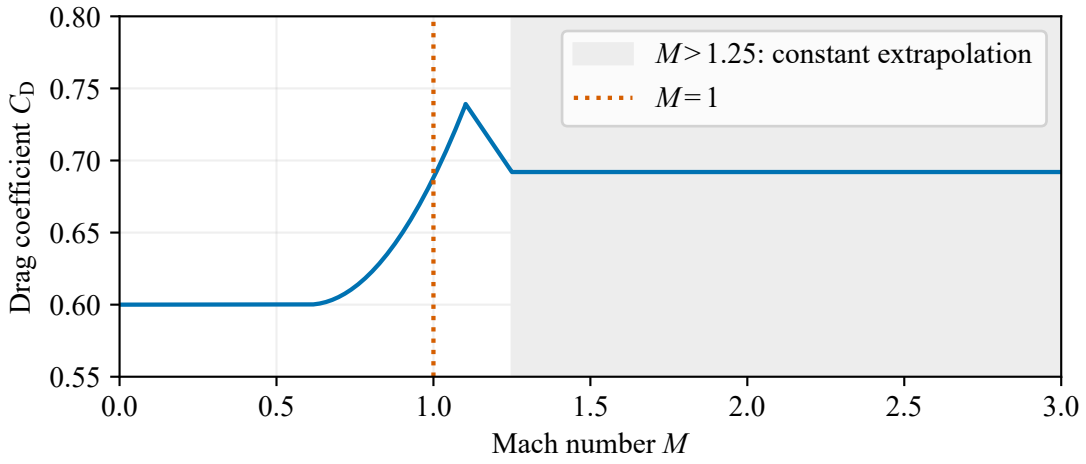


Figure 4: Adopted C_D–M relation; shading marks the constant extrapolation beyond the original data range.

With altitude positive upward, Newton's second law reads

$$m\frac{\mathrm{d}^2 h}{\mathrm{d}t^2} = -mg(h) + D.$$

The release conditions are $h(0) = h_0$ and $\dot{h}(0) = 0$. Defining downward speed as $v = -\dot{h}$ and applying the chain rule along the descent trajectory gives

$$v\frac{\mathrm{d}v}{\mathrm{d}h} = -g(h) + \frac{\rho(h)C_D(M)A}{2m}v^2. \tag{4.9}$$

Substitution of $v(h)$ into Eq. (4.7) then determines the drag profile $D(h)$.

4.2.2 Aerodynamic Heating

As the body moves rapidly relative to the air, the flow on its windward side decelerates, converting bulk kinetic energy into enthalpy. Near-wall viscous dissipation also raises the temperature. By the first law of thermodynamics, adiabatic deceleration to rest satisfies

$$c_p(T_0 - T) = \frac{v^2}{2}.$$

Here T_0 is the stagnation temperature. The constant-pressure specific heat c_p is the energy required to raise the temperature of unit mass of air by one degree at constant pressure; c_v in the specific-heat ratio $\gamma = c_p/c_v$ is the constant-volume specific heat. For an ideal gas with constant properties,

$$c_p = \frac{\gamma R}{\gamma - 1}.$$

With $\gamma = 1.4$ and the gas constant R defined above, $c_p = 1.005\,\mathrm{kJ/(kg\,K)}$. Viscous transport and thermal diffusion in the boundary layer allow near-wall air to recover part of the kinetic-energy temperature rise. We therefore introduce the recovery factor r_T:

$$c_p(T_{\mathrm{aw}} - T) = r_T \frac{v^2}{2}.$$

The Prandtl number Pr, the ratio of momentum diffusivity to thermal diffusivity, characterizes the relative strengths of these transport processes. According to a review of convective heat transfer [4, p. 58], the recovery factor for a turbulent boundary layer can be approximated by $r_T = \mathrm{Pr}^{1/3}$. Taking $\mathrm{Pr} = 0.71$ under the constant-property approximation for air yields $r_T = 0.8921$. The recovery-temperature profile is thus

$$T_{\mathrm{aw}}(h) = T(h) + r_T \frac{v(h)^2}{2c_p}. \tag{4.10}$$

Substituting the speed profile from Eq. (4.9) into Eq. (4.10) determines the recovery-temperature profile. Throughout this paper, T_{aw} denotes the adiabatic-wall reference temperature of the aerodynamic environment.

4.3 Attitude Dynamics

We approximate the mass distribution of the body and equipment by a uniform solid cylinder of length L and radius r. As described in Section 1.2, its axis follows the longitudinal axis of the body and lies in the horizontal plane, perpendicular to the vertical z axis. The geometry and angular coordinates are illustrated in Figure 5. We distinguish two rotational degrees of freedom: the pitch angle θ measures rotation about the y axis from the horizontal reference posture, and the flat-spin angle ψ measures rotation about the z axis.

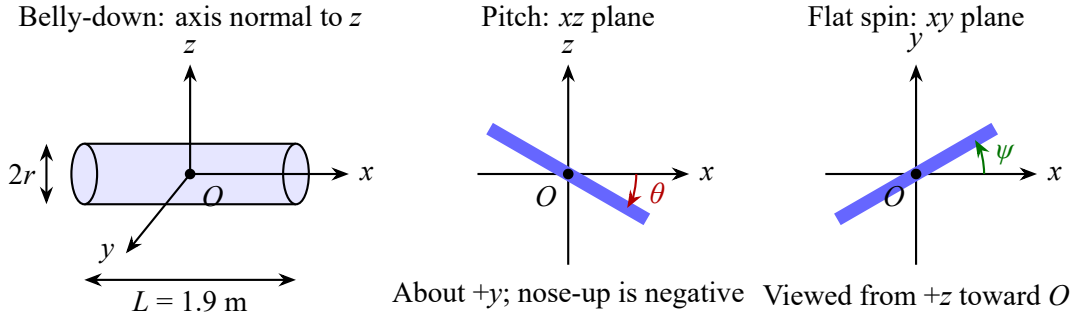


Figure 5: Uniform-cylinder approximation and angular coordinates.

Both reference rotation axes are perpendicular to the cylinder axis. The moment of inertia about the vertical axis is

$$I_z = \frac{m}{12}(L^2 + 3r^2).$$

In the reference posture, the pitch moment of inertia equals I_z. Using representative human dimensions $L = 1.9\,\text{m}$ and $r = 0.3\,\text{m}$ yields $I_z = 61.43\,\text{kg m}^2$. The distance from the head to the flat-spin axis is taken as $r_h = 0.5\,\text{m}$.

Pitch changes the angle between the body axis and the main airflow, thereby altering the windward pressure distribution. We assume that the reference posture is stable in pitch, so this change produces a restoring effect. Flat spin primarily changes the azimuth about the main airflow direction; approximately uniform vertical inflow does not provide the same angle-of-attack restoring mechanism, and persistent circumferential asymmetry can increase the spin rate. We therefore neglect the suppressed pitch perturbation and focus on flat spin.

Gravity acts through the center of mass and produces no moment about it. Following the treatment of yawing moments in the aerodynamics literature [5, p. 39], we separate the flat-spin torque into a driving bias term and a resisting damping term.

1. **Bias torque.** During rapid descent, the airflow is deflected and decelerated over the body and equipment, producing pressure and tangential forces. Asymmetry in shape or posture can leave a net torque:

$$\tau_{\text{bias}} = \frac{1}{2}\rho v^2 A L C_n.$$

   The positive direction is chosen to match the bias-driven rotation.

2. **Rotational damping.** Once the body rotates, a surface point at distance s from the axis acquires an additional tangential speed ωs. This changes the direction and magnitude of the local relative airflow. The resulting force change produces an opposing torque. For $|\omega|L/(2v) \ll 1$, the dependence of aerodynamic torque on angular speed is retained to first order:

$$\tau_{\text{damp}} = -\frac{1}{4}\rho v A L^2 C_{n\omega}\omega.$$

The parameters in these expressions are specified as follows:

1. **Descent speed.** Because the attitude-response timescale is much shorter than the free-fall timescale, v is treated as approximately constant.

2. **Area.** The value of A is the same as in Section 4.2.1.

3. **Bias coefficient.** Guided by the order of magnitude of yawing-moment coefficients in flight-test literature [6, p. 6], we take $C_n = 0.05$.

4. **Damping coefficient.** Using yaw-damping derivatives from an aircraft simulation study [5, p. 81] as a reference, we take the positive damping coefficient $C_{n\omega} = 1$.

Applying angular-momentum balance about the vertical axis through the center of mass yields a first-order linear equation for the flat-spin angular speed:

$$I_z \frac{d\omega}{d\tau} + \frac{1}{4}\rho v A L^2 C_{n\omega}\omega = \frac{1}{2}\rho v^2 A L C_n.$$

The analytical solution is

$$\omega(\tau) = \frac{2vC_n}{LC_{n\omega}}\left[1 - \exp\left(-\frac{\rho v A L^2 C_{n\omega}}{4I_z}\tau\right)\right]. \tag{4.11}$$

This exponential response reflects the competition between bias and damping. Substituting the local $\rho(h)$ and $v(h)$ at each altitude allows the rotational response over the same local time interval to be compared along the descent.

4.4 Parachute Descent

4.4.1 Inflation

After deployment, the packed canopy gradually inflates. Its effective area grows, increasing drag, while the falling speed decreases and reduces the quadratic speed factor in the drag. As illustrated in Figure 6, the forces are as follows:

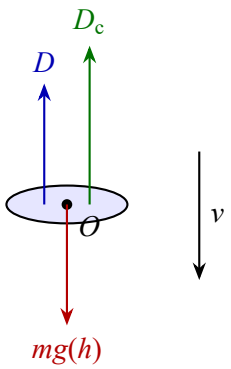


Figure 6: Forces during and after parachute inflation; D_c develops as the canopy fills.

1. **Gravity.** The downward force has magnitude $mg(h)$. The mass remains 190 kg, and gravitational acceleration follows Eq. (4.2).

2. **Body drag.** Equation (4.7) applies, with A equal to the calibrated effective area.

3. **Canopy drag.** This force acts upward, with magnitude

$$D_c = \frac{1}{2}\rho(h)v^2 B_c S(u). \tag{4.12}$$

The parameters and their basis are specified below.

1. **Parachute drag area.** The fully inflated drag area is $B_c = C_{D,c}A_c$, where A_c is the equivalent canopy reference area and $C_{D,c}$ is its drag coefficient. We take $C_{D,c} = 1.5$. In Section 5.1.2, B_c is calibrated to the sea-level descent speed, after which $A_c = B_c/C_{D,c}$.
2. **Inflation fraction.** Let t_o denote the deployment time and τ_c the inflation duration. With normalized inflation time $u = (t - t_o)/\tau_c$, $S(u)$ is the ratio of the instantaneous effective canopy drag area to its fully inflated value. We assume that this ratio starts at zero and reaches one at full inflation, with zero area-growth rate at both endpoints: $S(0) = 0$, $S(1) = 1$, $S'(0) = 0$, and $S'(1) = 0$. For $0 < u < 1$, the lowest-degree polynomial satisfying these four independent conditions yields

$$S(u) = \begin{cases} 0, & u \le 0, \\ 3u^2 - 2u^3, & 0 < u < 1, \\ 1, & u \ge 1. \end{cases}$$

Performance specifications from a parachute manufacturer describe a soft main-canopy opening over approximately 6 s to reduce opening shock. We adopt a slightly longer inflation duration, $\tau_c = 8$ s, as a design assumption and assess this choice using the calculated peak load in Section 6.4.

3. **Air density.** Equation (4.6) continues to apply, allowing density to vary with altitude during inflation.

Applying Newton's second law to the complete system yields

$$m\frac{\mathrm{d}^2 h}{\mathrm{d}t^2} = -mg(h) + D + D_c.$$

Using Eqs. (4.7) and (4.12), we obtain

$$v\frac{\mathrm{d}v}{\mathrm{d}h} = -g(h) + \frac{\rho(h)v^2}{2m}\left[C_D(M)A + B_c S(u)\right]. \tag{4.13}$$

4.4.2 Fully Inflated Descent and Landing

Once the canopy is fully inflated, $S = 1$. With $C_D = 0.6$ at low speed, Eq. (4.13) reduces to

$$v\frac{\mathrm{d}v}{\mathrm{d}h} = -g(h) + \frac{\rho(h)(0.6A + B_c)}{2m}v^2. \tag{4.14}$$

The descent speed therefore tends toward the local equilibrium speed:

$$v_{t,c}(h) = \sqrt{\frac{2mg(h)}{\rho(h)(0.6A + B_c)}}. \tag{4.15}$$

The values of A, B_c, and the gas properties are unchanged from the inflation stage. As altitude decreases, density increases and the equilibrium speed slowly falls; the actual descent speed gradually follows this variation. The landing speed v_{land} is therefore approximately $v_{t,c}(0)$.

5 Model Calibration and Results

5.1 Calibration and Model Checks

Two parameters in the preceding models remain to be determined: the effective body frontal area A and the parachute drag area B_c. Here we use reference data to calibrate both parameters and to assess the representation of the atmosphere and the descent process.

5.1.1 Atmosphere Check

We select records for stations 72364 and 72365 from the public upper-air sounding archive for the two stations to assess the temperature model in Eq. (4.3) and the density model in Eq. (4.6). The reference data comprise eight soundings: both stations at 12 UTC on March 15, July 25, and October 14, 2012, and at 00 UTC on October 15. Observed density is calculated from measured pressure and temperature as $p/(RT)$. Geopotential altitude is converted to geometric altitude using Eq. (4.1) before comparison.

As shown in Figure 7, the October temperature and density profiles at both stations broadly follow the mean vertical structure of the standard atmosphere. Across all eight soundings at geopotential altitudes of 10, 20, and 25 km, the maximum absolute relative density deviation is 8.66%.

5.1.2 Parachute Drag Area

According to a parachute manufacturer's performance specifications, the reference main canopy achieves a sea-level descent speed below 18 ft/s, approximately 5.49 m/s, at a total load of about 181 kg. Using this speed as a reference, we choose $v_d = 5.5$ m/s as the design sea-level

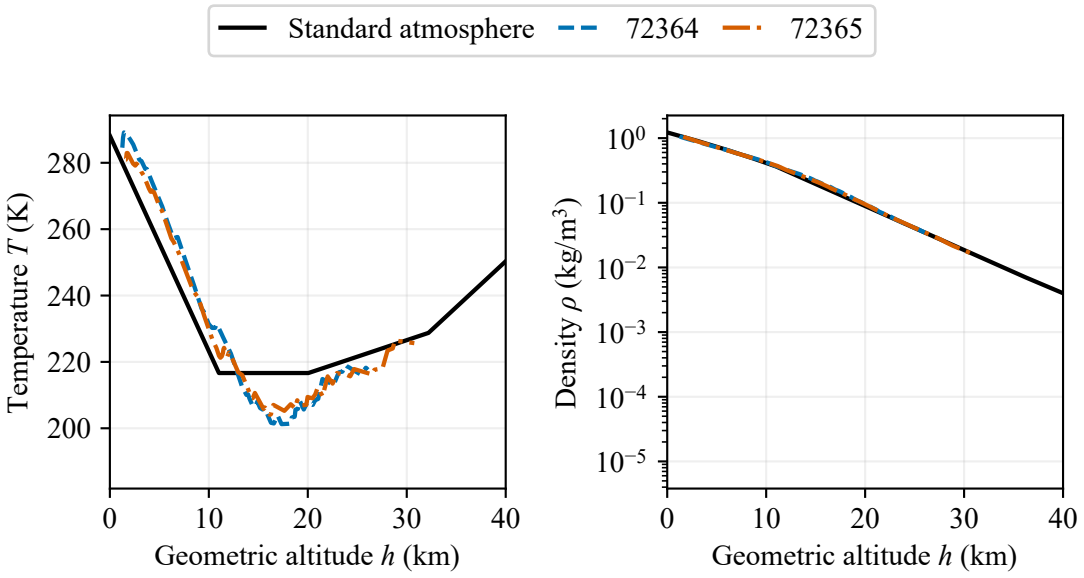


Figure 7: Comparison of the standard atmosphere with two soundings at 12 UTC on October 14, 2012.

equilibrium speed for the specified 190 kg system and determine the required parachute drag area.

Setting $h = 0$ and $v_{t,c}(0) = v_d$ in Eq. (4.15) and rearranging yields

$$B_c = \frac{2mg_0}{\rho(0)v_d^2} - 0.6A.$$

With sea-level density $\rho(0) = 1.225\,\mathrm{kg/m^3}$ and $A = 1.239\,\mathrm{m^2}$ calibrated from the three peak speeds in the next subsection, we obtain $B_c = 99.82\,\mathrm{m^2}$.

5.1.3 Joint Frontal-Area Calibration

In 2012, Austrian space diver Felix Baumgartner ascended in a high-altitude balloon capsule and descended in a pressure suit. He completed two space-diving tests on March 15 and July 25, followed by the program's main space dive on October 14. We compile the release altitudes and peak speeds of all three dives from the mission report [1]. These observations describe the speed response from different initial altitudes, allowing the unknown effective frontal area A to be jointly determined once the drag-coefficient relation is specified.

Following the mass used in the aerodynamic study [2], we take $m_{\mathrm{ref}} = 121.2\,\mathrm{kg}$. Each dive starts from its own release altitude with $v(0) = 0$. We integrate the equivalent time-domain form of Eq. (4.9), together with Eqs. (4.6) and (4.8), down to the deployment altitude and extract the maximum speed of each trajectory, $v_{\mathrm{max},i}^{\mathrm{model}}(A)$. We then minimize the sum of squared relative peak-speed residuals:

$$A^* = \underset{0.3\le A\le 3.0\ \mathrm{m^2}}{\arg\min} \sum_{i=1}^{3} \left[\frac{v_{\mathrm{max},i}^{\mathrm{model}}(A)}{v_{\mathrm{max},i}^{\mathrm{obs}}} - 1 \right]^2.$$

The fitted area is $A^* = 1.239\,\mathrm{m}^2$.

As shown in Figure 8, the fitted peak speeds agree closely with all three observations.

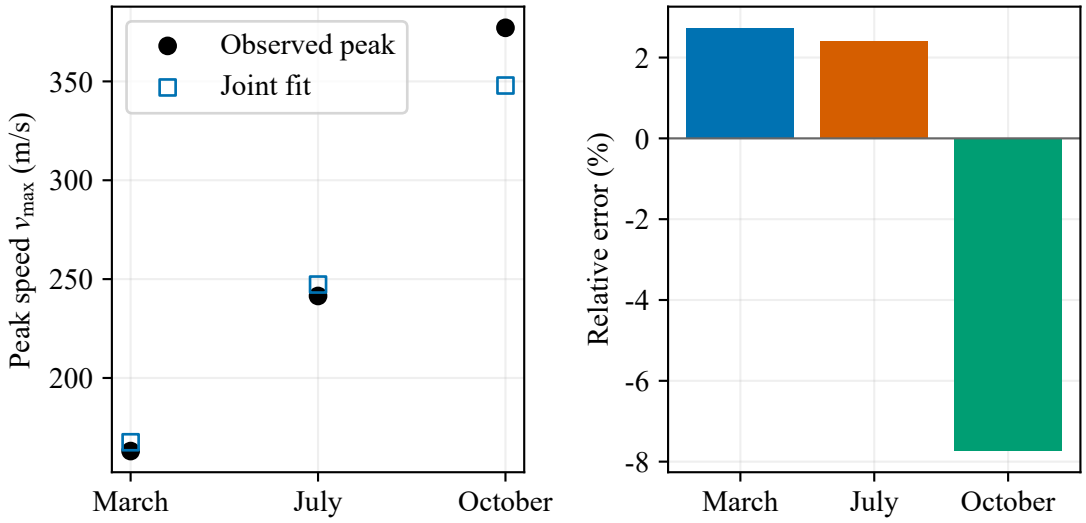


Figure 8: Observed and jointly fitted peak speeds, with relative errors for the three dives.

5.1.4 October Speed Profile

During the main space dive on October 14, 2012, an onboard GPS unit recorded the descent trajectory. An aerodynamic study [2] used these measurements to analyze human-body aerodynamics; its speed–altitude curve provides an observational basis for comparing the full descent. We compare the simulated profile with this measured curve to assess whether calibration using peak speeds also captures the complete motion.

As shown in Figure 9, the simulated and observed curves exhibit similar acceleration and deceleration trends.

5.2 Descent Characteristics

For the 190 kg system, we consider release from rest at $h_0 = 40, 50, 60, 70, 80\,\mathrm{km}$. The descent trajectories follow the equivalent time-domain form of Eq. (4.9); drag and recovery temperature then follow Eqs. (4.7) and (4.10). The predicted speed, drag, and temperature profiles are shown in Figures 10, 11, and 13, respectively. They reveal the following three physical features.

5.2.1 Transonic Drag Rise

As seen in Figure 10, a higher release altitude provides a longer acceleration distance in thin air and generally produces a higher peak speed. Near the speed of sound, the drag coefficient

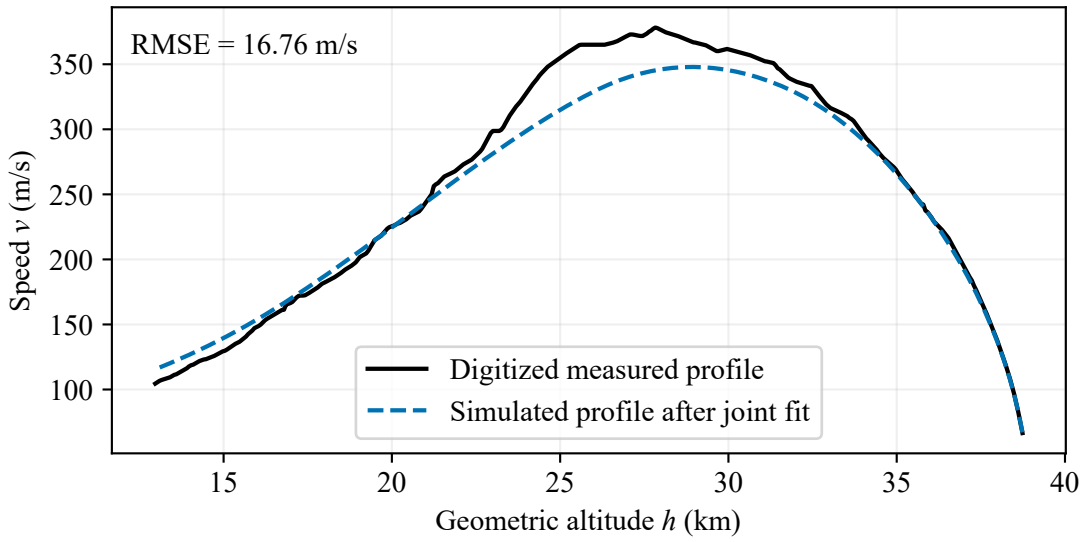


Figure 9: Simulated October speed–altitude profile compared with digitized values from the published measured curve.

in Eq. (4.8) rises from its low-speed plateau to a transonic peak. This is the additional drag associated with compressibility, commonly described as the sound-barrier effect.

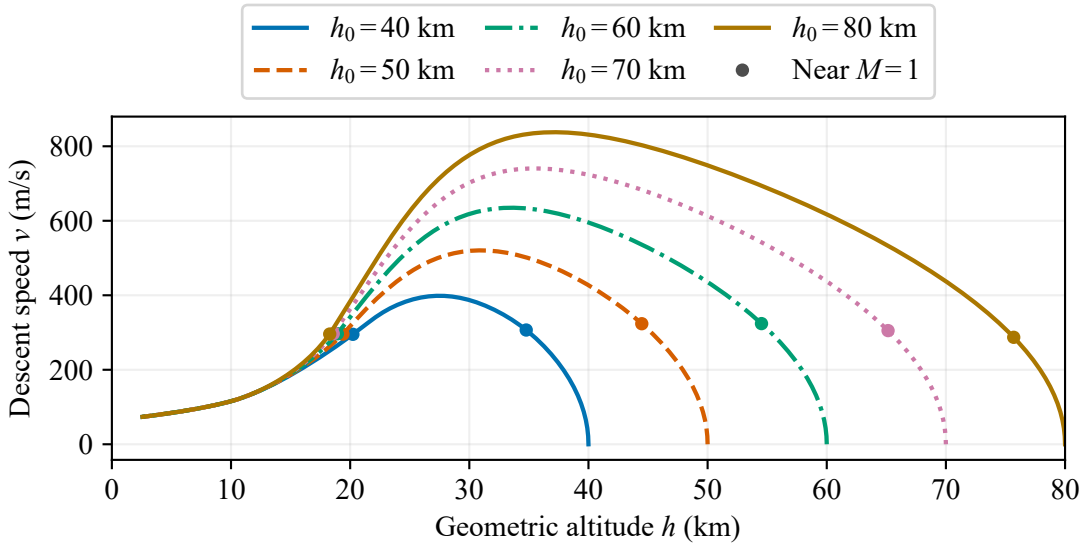


Figure 10: Speed–altitude profiles for different release altitudes; dots mark locations near $M = 1$.

5.2.2 Terminal-Speed Convergence

The local terminal speed in the reference posture follows from force balance:

$$\frac{1}{2}\rho(h)C_D(v_t/c)Av_t^2 = mg(h).$$

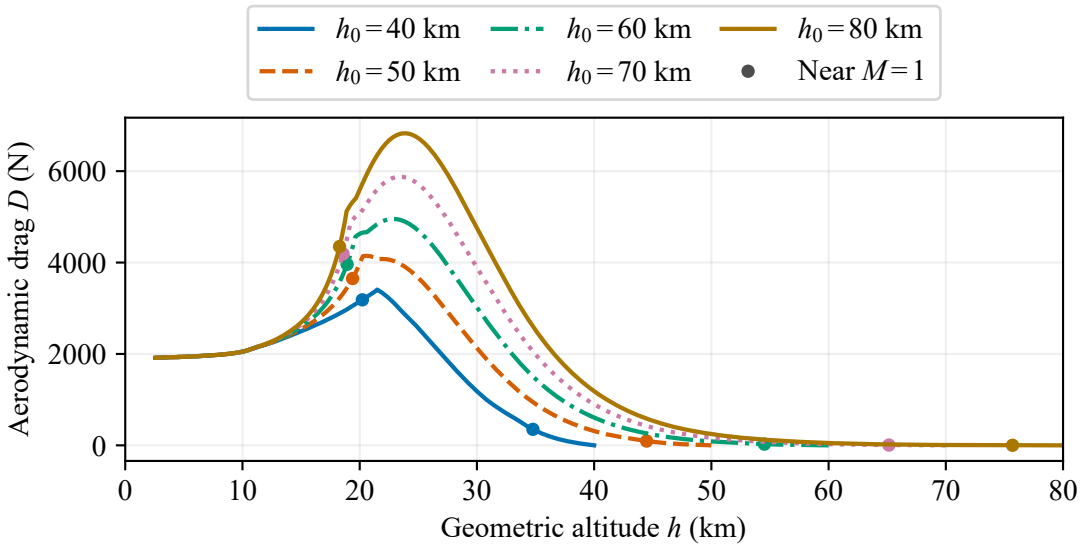


Figure 11: Aerodynamic drag–altitude profiles for different release altitudes; dots mark locations near $M = 1$.

In the low-altitude subsonic region, $C_D = 0.6$, so

$$v_t(h) = \sqrt{\frac{2mg(h)}{0.6\rho(h)A}}. \tag{5.1}$$

Equation (5.1) shows that, for specified mass, effective area, and local atmospheric conditions, terminal speed does not depend on release altitude h_0. The calculated values are 63.97 m/s at sea level and 82.43 m/s at 5 km.

This behavior is also apparent in Figure 12: the low-altitude speed profiles from releases at 40–80 km gradually converge toward the same local equilibrium curve.

5.2.3 Recovery-Temperature Profile

As seen in Figure 13, recovery temperature first rises markedly during acceleration and then falls as the diver decelerates at lower altitudes.

When speed reaches its maximum, the ambient atmospheric temperature may continue to change during descent. Equation (4.10) therefore implies that the recovery-temperature peak need not coincide with the speed peak.

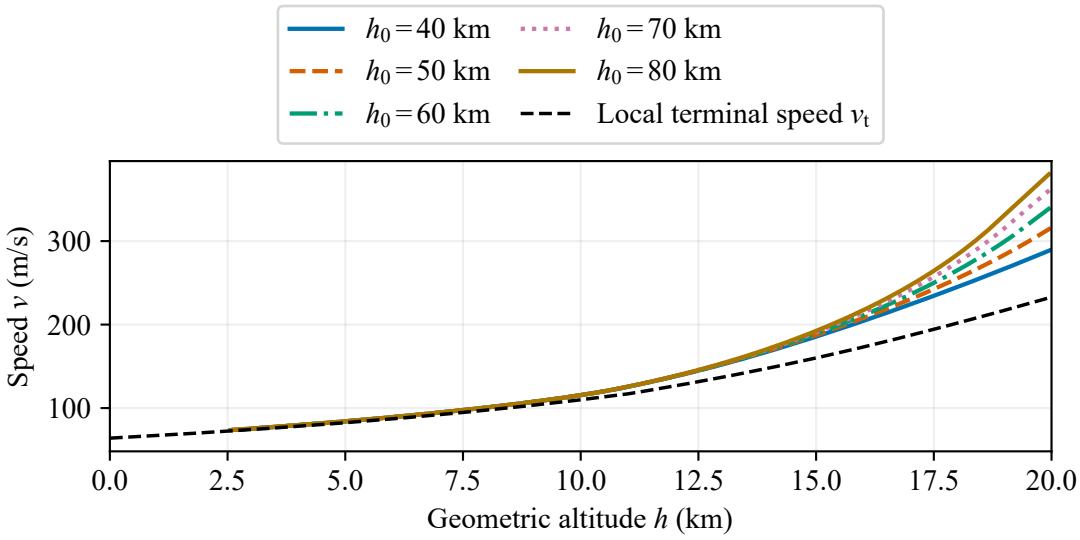


Figure 12: Speed profiles and local free-fall terminal speed over 0–20 km. Trajectories from different release altitudes converge during descent and end at parachute deployment.

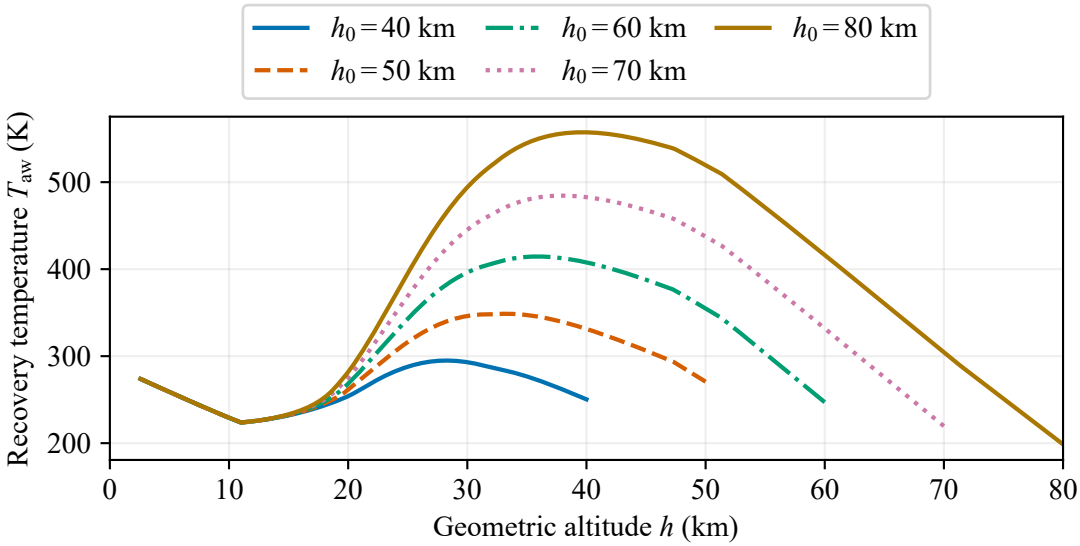


Figure 13: Recovery-temperature profiles for different release altitudes.

6 Hazard Assessment

Section 5 shows that a higher release altitude generally leads to a higher peak speed and greater aerodynamic heating. Entry into denser air also produces substantial drag and aerodynamic torque. The main hazards during descent can therefore be grouped into free-fall loads, high temperature, and rotation. Meanwhile, low-altitude speed convergence makes the state at parachute deployment insensitive to sufficiently high release altitudes, and the equilibrium speed under a fully inflated canopy depends primarily on local density and canopy parameters.

We first examine the three main hazards, then use the low-altitude behavior to assess the selected deployment and landing scheme.

6.1 Free-Fall Loads

Drag decelerates the body while imposing nongravitational loads on the diver and equipment. The free-fall hazard is therefore assessed using aerodynamic specific force. Define

$$n_D(h) = \frac{D(h)}{mg_0}.$$

We adopt the peak-load criterion

$$\max_{\text{free fall}} n_D \le 5. \tag{6.1}$$

The net acceleration $\dot{v} = g - D/m$ differs from the aerodynamic specific force D/m experienced by the body. In particular, the latter is small during the initial near-ballistic fall but may exceed g_0 during deceleration.

In a belly-down posture, the dominant aerodynamic force acts approximately along the chest–back direction, so the human G_x direction in a crewed-spaceflight standard provides an order-of-magnitude reference. The standard [7, Sec. 6.5.1, Fig. 4] indicates that load limits depend on duration and physiological condition. We choose a load-factor threshold of 5 for this model.

We calculate the free-fall load using Eq. (4.9) and determine its altitude boundary using Eq. (6.1). As shown in Figure 14, the peak load rises with release altitude. Over 0.1–86 km, its maximum is 3.975 g_0, below the design limit. To estimate the individual upper altitude for this criterion, only this subsection continuously extrapolates the temperature gradient, hydrostatic pressure, and density relations of the final atmospheric layer to a geometric altitude of 110 km. The resulting free-fall load boundary is 105.58 km. In this extrapolated case, the peak of 5 g_0 occurs 134.28 s after release at a geometric altitude of 24.46 km.

6.2 Thermal Hazard

Aerodynamic heating may affect the outer suit, visor, and seals. The quantity calculated here is the recovery temperature in Eq. (4.10), for which we impose the screening condition

$$\max_{\text{free fall}} T_{\text{aw}}(h) \le T_{\text{lim}}.$$

Published spacesuit environmental information describes temperatures during a spacewalk ranging from about −250 °F to about 250 °F in sunlight, or approximately 116–394 K. Using the upper end of this environmental range as a reference and rounding downward, we select $T_{\text{lim}} = 390$ K as the recovery-temperature screening value.

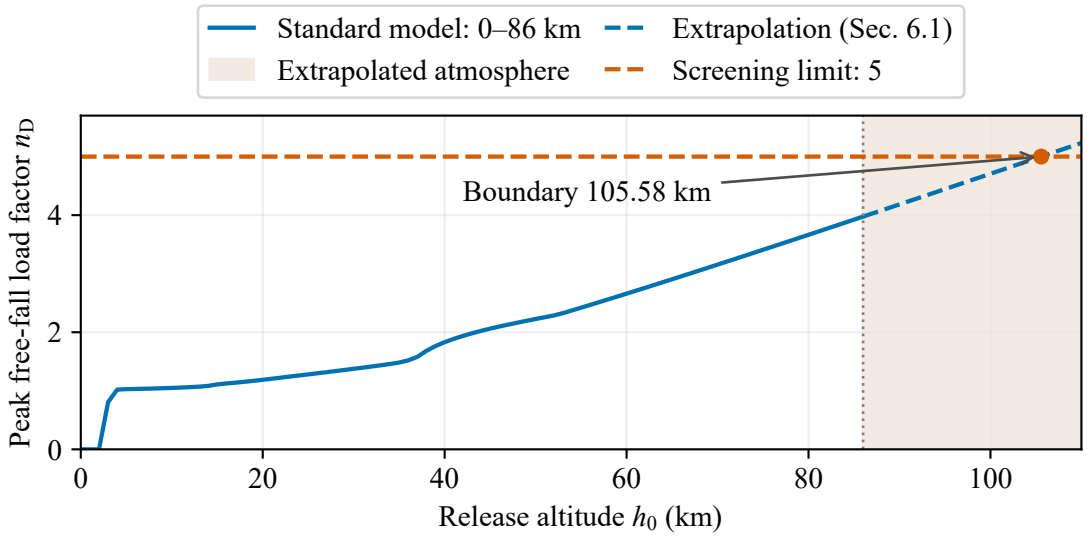


Figure 14: Peak free-fall load and screening limit. Shading and the dashed curve above 86 km denote the atmospheric extrapolation used only in this subsection.

As shown in Figure 15, peak recovery temperature increases with release altitude and intersects the screening limit at 56.36 km. In this limiting case, the recovery temperature reaches 390 K at 67.45 s after release and a geometric altitude of 35.02 km.

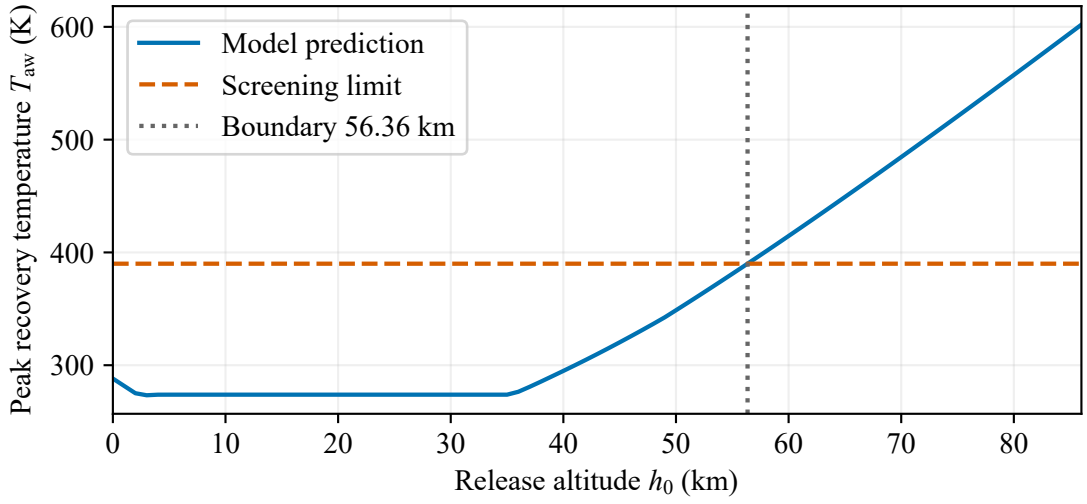


Figure 15: Peak recovery temperature versus release altitude. The intersection identifies the altitude boundary for the selected thermal design condition.

6.3 Flat-Spin Hazard

Flat spin subjects the head to radial acceleration, potentially affecting blood distribution and attitude control. The linear damping model in Section 4 describes local perturbations whose rotational speed is small relative to the main airflow. The diver must therefore detect the spin

promptly and adjust posture to reduce aerodynamic asymmetry. We assume a total of 1 s from the onset of the bias through detection to completion of the adjustment. Consistent with Section 6.1, we select $5\,g_0$ as the head radial-acceleration threshold for this brief response. With $r_h = 0.5\,\mathrm{m}$ and Eq. (4.11), we calculate the following separately at every position along the trajectory:

$$n_r(h) = \frac{r_h[\omega(1\,\mathrm{s}; h)]^2}{g_0}.$$

The entire free-fall segment must satisfy

$$\max_{\text{free fall}} n_r(h) \le 5. \tag{6.2}$$

As shown in Figure 16, the maximum local 1 s radial acceleration increases with release altitude and reaches the limit in Eq. (6.2) at $h_0 = 66.28\,\mathrm{km}$.

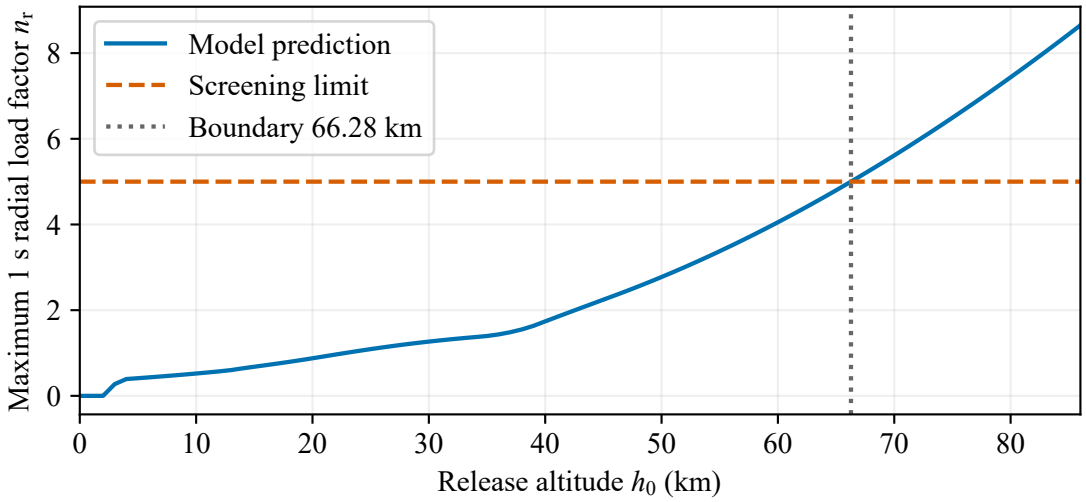


Figure 16: Maximum local 1 s radial acceleration versus release altitude and the flat-spin screening boundary.

6.4 Parachute-Opening Loads

Opening loads are transmitted to the body through the harness and are measured by the total aerodynamic specific force:

$$n_o = \frac{D + D_c}{mg_0}.$$

We use the same threshold as in Section 6.1:

$$\max_{\text{inflation and canopy descent}} n_o \le 5. \tag{6.3}$$

The low-altitude convergence in Section 5 implies that, at a fixed deployment altitude and with fixed canopy parameters, the deployment states from sufficiently high releases approach one another. Satisfaction of Eq. (6.3) therefore depends primarily on the selected low-altitude deployment state and canopy parameters. For releases at 40–80 km, the spread in deployment speed is less than 1.00×10^{-4} m/s, and the spread in peak opening load is less than $1.00\times10^{-4}\,g_0$. These differences are very small.

Following the deployment altitude in the flight study and mission records [1], [2], we take $h_o = 2567$ m and apply Eq. (4.13). As shown in Figure 17, the peak opening load is $4.660\,g_0$, below the threshold. This low-altitude deployment scheme therefore does not limit the release altitude.

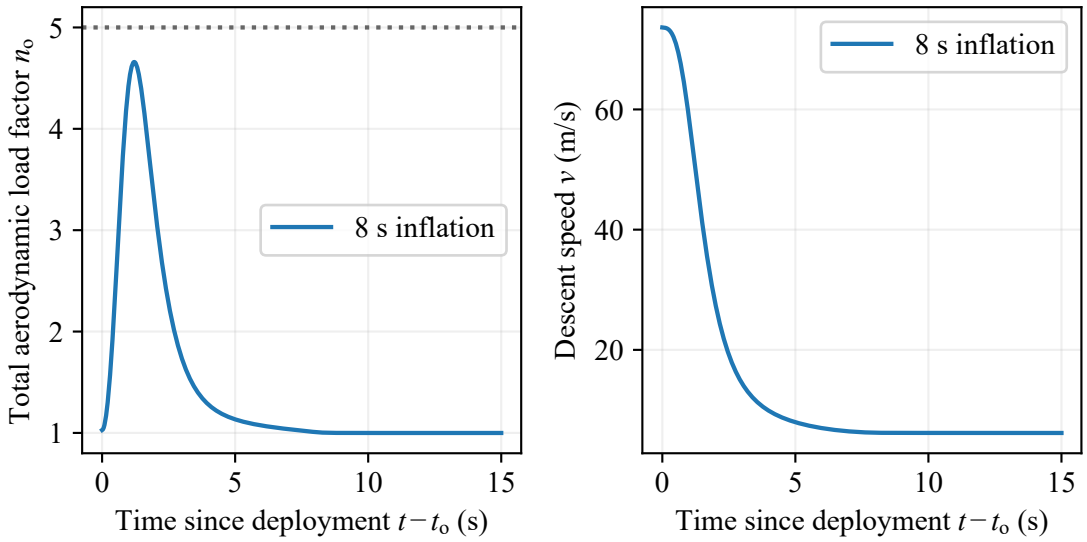


Figure 17: Load and speed during the selected 8 s inflation, using the deployment state at the combined altitude boundary.

6.5 Landing

We assess landing using the vertical speed immediately before ground contact. With reference to sea-level descent speeds in parachute manufacturer specifications, we adopt

$$v_{\text{land}} \le 6\ \text{m/s}.$$

As in Section 6.4, the local equilibrium speed under a fully inflated canopy is essentially independent of release altitude. Keeping the 2567 m deployment altitude, Eq. (4.14) yields a landing speed of 5.500 m/s at the combined boundary, below 6 m/s. Landing speed therefore imposes no maximum release altitude.

7 Conclusions

For the 190 kg system, the model shows that transonic drag rise, low-altitude speed convergence, and aerodynamic heating govern free fall, the deployment state, and the thermal hazard, respectively. The maximum altitudes under all implemented conditions are summarized in Table 2.

Table 2: Individual altitude boundaries and the combined result.

Constraint	Criterion	Basis	Maximum altitude (km)	Governing
Free-fall load	$\max n_D \leq 5$	Spaceflight reference; assumption	105.58 (extrapolated)	No
Recovery temperature	$\max T_{\mathrm{aw}} \leq 390\,\mathrm{K}$	Thermal environment; assumption	56.36	Yes
1 s radial load	$\max n_r \leq 5$	1 s response; assumption	66.28	No
Opening load	$\max n_o \leq 5$	Load assumption	No boundary within 0–86 km	No
Landing speed	$v_{\mathrm{land}} \leq 6\,\mathrm{m/s}$	Canopy reference; assumption	No boundary within 0–86 km	No
Combined	Intersection of five feasible sets	Selected design scenario	56.36	Thermal

Under the assumptions of this study, the combined maximum release altitude is **56.36 km**. At this altitude, peak speed is 594.4 m/s and peak Mach number is 1.960. Peak free-fall and opening loads are 2.484 g_0 and 4.660 g_0, respectively; landing speed is 5.500 m/s, and the complete descent lasts approximately 653.3 s. The 390 K recovery-temperature design condition determines this altitude.

8 Model Evaluation

8.1 Strengths

The model connects the standard atmosphere, translation, aerodynamic heating indicator, attitude, and parachute descent through force balance, energy conservation, and angular-momentum balance. Calibration and prediction use the same Mach-dependent drag relation. Fitting three peak speeds and comparing the full October profile provide evidence at different levels of detail.

8.2 Limitations

A single effective area and a common reference mass cannot fully distinguish differences in posture and equipment among the three dives. October data are used in peak-speed calibration, and the drag coefficient is derived from the same flight, so the profile comparison is not an independent predictive validation. The constant extrapolation of drag for $M > 1.25$ lies outside the original empirical range, whereas the central case reaches a peak Mach number of 1.960. This limits confidence in the high-altitude predictions.

The cylindrical inertia and torque coefficients are simplified design assumptions, and transferring the order of magnitude of aircraft aerodynamic derivatives to the human body lacks direct human measurements. Neglecting pitch requires a stable reference posture and small perturbations. The local flat-spin solution freezes density and speed and assumes that adjustment is completed within 1 s of bias onset; it does not represent cumulative rotation or the complete active-control process. Recovery temperature remains an aerodynamic-environment indicator. The 390 K, load, and landing screening values require further correspondence with specific equipment and human conditions.

References

[1] Red Bull Stratos, *Summary Report: Findings of the Red Bull Stratos Scientific Summit*, California Science Center, Los Angeles, CA, USA, 2013.

[2] M. Guerster and U. Walter, "Aerodynamics of a highly irregular body at transonic speeds—Analysis of STRATOS flight data," *PLoS ONE*, vol. 12, no. 12, Dec. 2017, Art. no. e0187798, doi: 10.1371/journal.pone.0187798.

[3] NOAA, NASA, and USAF, *U.S. Standard Atmosphere, 1976*. Washington, DC, USA: U.S. Government Printing Office, Oct. 1976, Rep. NASA-TM-X-74335.

[4] S. Chung and F. L. Staffanson, "Survey of literature on convective heat transfer coefficients and recovery factors for high atmosphere thermometry," Univ. Utah, Salt Lake City, UT, USA, NASA-CR-138747, Jul. 1973.

[5] L. T. Nguyen, M. E. Ogburn, W. P. Gilbert, K. S. Kibler, P. W. Brown, and P. L. Deal, "Simulator study of stall/post-stall characteristics of a fighter airplane with relaxed longitudinal static stability," NASA Langley Research Center, Hampton, VA, USA, NASA-TP-1538, Dec. 1979.

[6] B. R. Cobleigh, "High-angle-of-attack yawing moment asymmetry of the X-31 aircraft from flight test," NASA Dryden Flight Research Center, Edwards, CA, USA, NASA-CR-186030, 1994.

[7] *NASA Space Flight Human-System Standard, Volume 2: Human Factors, Habitability, and Environmental Health*, NASA-STD-3001, vol. 2, rev. C, NASA, Apr. 8, 2022.

A Solution Code

```
"""Standalone Python: print three height limits (km); no third-party imports."""
import math
from bisect import bisect_right

# SI units. US Standard Atmosphere 1976.
RE, G0, R, GAMMA = 6356766., 9.80665, 287.05287, 1.4
CP, RECOVERY = GAMMA * R / (GAMMA - 1), .71 ** (1 / 3)
MASS, REFERENCE_MASS, OPENING = 190., 121.2, 2566.8
INFLATION, DESIGN_SPEED, CANOPY_CD = 8., 5.5, 1.5
LENGTH, RADIUS, HEAD_RADIUS = 1.9, .3, .5
CN, CNW, RESPONSE = .05, 1., 1.
INERTIA = MASS * (LENGTH ** 2 + 3 * RADIUS ** 2) / 12
HB = (0., 11000., 20000., 32000., 47000., 51000., 71000., 84852.)
LAPSE = (-.0065, 0., .001, .0028, 0., -.0028, -.002)
TB, PB = [288.15], [101325.]
for i in range(7):
    dz = HB[i + 1] - HB[i]
    temp = TB[i] + LAPSE[i] * dz
    factor = (math.exp(-G0 * dz / (R * TB[i])) if LAPSE[i] == 0
              else (temp / TB[i]) ** (-G0 / (R * LAPSE[i])))
    TB.append(temp)
    PB.append(PB[i] * factor)

# Stratos Scientific Summit Summary Report, 2013: March, July, October.
# Each pair is release height (m), observed peak speed (m/s).
FLIGHTS = ((21828.3, 163.0333), (29610., 241.4722), (38969.4, 377.1))

def atmosphere(h, load_extension=False):
    top = 110000. if load_extension else 86000.
    if not math.isfinite(h) or h > top + .01:
        raise ValueError("Height outside atmosphere domain")
    h = max(h, 0.)
    z = RE * h / (RE + h)
    i = min(6, bisect_right(HB, z) - 1)
    dz = z - HB[i]
    temp = TB[i] + LAPSE[i] * dz
    factor = (math.exp(-G0 * dz / (R * TB[i])) if LAPSE[i] == 0
              else (temp / TB[i]) ** (-G0 / (R * LAPSE[i])))
    return temp, PB[i] * factor / (R * temp), math.sqrt(GAMMA * R * temp)

def gravity(h):
    return G0 * (RE / (RE + max(h, 0.))) ** 2

def drag_coefficient(mach):
```

```
    mach = abs(mach)
    if mach <= .6:
        return .6
    if mach <= 1.1:
        return .6 + .55 * (mach - .6) ** 2
    if mach <= 1.25:
        return .74 - .32 * (mach - 1.1)
    return .692

def minimize(function, left, right, tolerance=1e-7):
    # Golden-section minimization in a bracket.
    ratio = (math.sqrt(5.) - 1.) / 2
    a, b = right - ratio * (right - left), left + ratio * (right - left)
    fa, fb = function(a), function(b)
    while right - left > tolerance:
        if fa < fb:
            right, b, fb = b, a, fa
            a = right - ratio * (right - left)
            fa = function(a)
        else:
            left, a, fa = a, b, fb
            b = left + ratio * (right - left)
            fb = function(b)
    return (a, fa) if fa < fb else (b, fb)

def root(function, left, right, tolerance=.01):
    # Bisection, retaining a sign-changing bracket.
    fl, fr = function(left), function(right)
    if fl == 0:
        return left
    if fr == 0:
        return right
    if fl * fr > 0:
        raise ValueError("Root is not bracketed")
    while right - left > tolerance:
        middle = (left + right) / 2
        fm = function(middle)
        if fm == 0:
            return middle
        if fl * fm <= 0:
            right = middle
        else:
            left, fl = middle, fm
    return (left + right) / 2

def rk4(rhs, t, y, dt):
    # Classical fourth-order Runge-Kutta step for (height, speed).
    k1 = rhs(t, y)
```

```
    k2 = rhs(t + dt / 2, tuple(y[j] + dt * k1[j] / 2 for j in (0, 1)))
    k3 = rhs(t + dt / 2, tuple(y[j] + dt * k2[j] / 2 for j in (0, 1)))
    k4 = rhs(t + dt, tuple(y[j] + dt * k3[j] for j in (0, 1)))
    return tuple(y[j] + dt * (k1[j] + 2*k2[j] + 2*k3[j] + k4[j]) / 6
                 for j in (0, 1))

def interpolate(y0, y1, f0, f1, dt, u):
    # Cubic Hermite dense output from endpoint states and derivatives.
    a, b = 2*u**3 - 3*u**2 + 1, u**3 - 2*u**2 + u
    c, d = -2*u**3 + 3*u**2, u**3 - u**2
    return tuple(a*y0[j] + b*dt*f0[j] + c*y1[j] + d*dt*f1[j]
                 for j in (0, 1))

class Trajectory:
    def __init__(self, rhs, initial):
        self.times = [0.]
        self.states = [initial]
        self.slopes = [rhs(0., initial)]

    def at(self, t):
        if not self.times[0] <= t <= self.times[-1]:
            raise ValueError("Time outside trajectory")
        i = min(bisect_right(self.times, t) - 1, len(self.times) - 2)
        dt = self.times[i + 1] - self.times[i]
        return interpolate(self.states[i], self.states[i + 1],
                           self.slopes[i], self.slopes[i + 1], dt,
                           (t - self.times[i]) / dt)

def integrate(rhs, initial, stop_height, max_step=.1):
    # Adaptive RK4 step doubling; accept the Richardson-corrected state.
    path = Trajectory(rhs, initial)
    dt, t, y = min(.05, max_step), 0., initial
    while t < 3000.:
        dt = min(dt, max_step, 3000. - t)
        coarse = rk4(rhs, t, y, dt)
        half = rk4(rhs, t, y, dt / 2)
        fine = rk4(rhs, t + dt / 2, half, dt / 2)
        correction = tuple((fine[j] - coarse[j]) / 15 for j in (0, 1))
        error = max(abs(correction[j]) /
                    (1e-12 + 1e-11 * max(abs(y[j]), abs(fine[j])))
                    for j in (0, 1))
        if not math.isfinite(error):
            raise RuntimeError("Nonfinite integration state")
        factor = 2. if error == 0 else min(2., max(.2, .9 *
        ↪  error**(-.2)))
        if error > 1:
            dt *= factor
            if dt < 1e-8:
```

```
                raise RuntimeError("Integration step too small")
            continue
        new = tuple(fine[j] + correction[j] for j in (0, 1))
        slope = rhs(t + dt, new)
        if new[0] <= stop_height:
            f0 = path.slopes[-1]
            fraction = root(lambda u: interpolate(y, new, f0, slope, dt,
             ↪  u)[0]
                            - stop_height, 0., 1., 1e-11)
            final = interpolate(y, new, f0, slope, dt, fraction)
            path.times.append(t + dt * fraction)
            path.states.append((stop_height, final[1]))
            path.slopes.append(rhs(path.times[-1], path.states[-1]))
            return path
        t, y = t + dt, new
        path.times.append(t)
        path.states.append(y)
        path.slopes.append(slope)
        dt *= factor
    raise RuntimeError("Height event not reached")

def freefall(h0, mass, area, load_extension=False):
    atmosphere(h0, load_extension)
    if h0 <= OPENING:
        raise ValueError("Release must precede parachute opening")
    def rhs(t, y):
        h, v = y
        _, rho, sound = atmosphere(h, load_extension)
        drag = .5 * rho * drag_coefficient(v / sound) * area * v *
         ↪  abs(v)
        return (-v, gravity(h) - drag / mass)
    return integrate(rhs, (h0, 0.), OPENING)

def calibrate():
    def peak(area, height):
        path = freefall(height, REFERENCE_MASS, area)
        k = max(range(len(path.states)), key=lambda i:
         ↪  path.states[i][1])
        lo, hi = max(0, k - 1), min(len(path.times) - 1, k + 1)
        return -minimize(lambda t: -path.at(t)[1],
                         path.times[lo], path.times[hi])[1]
    def objective(area):
        return sum((peak(area, h) / v - 1)**2 for h, v in FLIGHTS)
    return minimize(objective, .3, 3., 1e-9)[0]

def metric(h, v, area, kind):
    temp, rho, sound = atmosphere(h, load_extension=(kind == "load"))
    if kind == "load":
```

```
        return .5 * rho * drag_coefficient(v / sound) * area * v**2 /
        ↪ (MASS * G0)
    if kind == "heat":
        return temp + RECOVERY * v**2 / (2 * CP)
    if kind == "spin":
        rate = rho * v * area * LENGTH**2 * CNW / (4 * INERTIA)
        omega = 2 * v * CN / (LENGTH * CNW) * (-math.expm1(-rate *
        ↪ RESPONSE))
        return HEAD_RADIUS * omega**2 / G0
    raise ValueError("Unknown constraint")

def peak_value(path, function):
    count = max(500, int(path.times[-1] / .25))
    times = [path.times[-1] * i / count for i in range(count + 1)]
    times[-1] = path.times[-1]
    values = [function(t) for t in times]
    best = max(values)
    for k in range(1, count):
        if values[k] >= values[k-1] and values[k] >= values[k+1]:
            best = max(best, -minimize(lambda t: -function(t),
                                       times[k-1], times[k+1])[1])
    return best

def maximum(h0, area, kind):
    path = freefall(h0, MASS, area, load_extension=(kind == "load"))
    return peak_value(path, lambda t: metric(*path.at(t), area, kind))

def height_limit(area, kind, limit):
    top = 110000 if kind == "load" else 86000
    high = float(top)
    if maximum(high, area, kind) <= limit:
        raise RuntimeError("No finite boundary inside model domain")
    for low in range(top - 1000, int(OPENING), -1000):
        if maximum(float(low), area, kind) <= limit:
            return root(lambda h: maximum(h, area, kind) - limit,
                        float(low), high)
        high = float(low)
    raise RuntimeError("No feasible bracket found")

def canopy_descent(h_open, v_open, area):
    # Fully inflated drag area, calibrated from sea-level equilibrium.
    rho0 = atmosphere(0.)[1]
    bc = 2 * MASS * G0 / (rho0 * DESIGN_SPEED**2) - .6 * area
    def forces(t, h, v):
        _, rho, sound = atmosphere(h)
        u = min(1., max(0., t / INFLATION))
        fraction = u*u*(3 - 2*u)
```

```
        body = .5 * rho * drag_coefficient(v / sound) * area * v *
        ↪ abs(v)
        canopy = .5 * rho * bc * fraction * v * abs(v)
        return body + canopy
    def rhs(t, y):
        h, v = y
        return (-v, gravity(h) - forces(t, h, v) / MASS)
    # One continuous ODE covers inflation, full canopy and landing at
    ↪ h=0.
    path = integrate(rhs, (h_open, v_open), 0., max_step=.25)
    load = peak_value(path, lambda t: forces(t, *path.at(t)) / (MASS *
    ↪ G0))
    return {"trajectory": path, "drag_area_m2": bc,
            "canopy_area_m2": bc / CANOPY_CD,
            "opening_speed_mps": v_open, "opening_load_g": load,
            "landing_speed_mps": path.states[-1][1],
            "inflation_completed": path.times[-1] >= INFLATION,
            "full_inflation_height_m": path.at(INFLATION)[0]
            if path.times[-1] >= INFLATION else None}

def complete_descent(h0, area):
    free = freefall(h0, MASS, area)
    parachute = canopy_descent(*free.states[-1], area)
    return {"freefall": free, "parachute": parachute,
            "total_time_s": free.times[-1] +
            ↪ parachute["trajectory"].times[-1]}

def main():
    area = calibrate()
    heights = [height_limit(area, kind, limit) for kind, limit in
               (("load", 5.), ("heat", 390.), ("spin", 5.))]
    descent = complete_descent(min(heights), area)
    for h in heights:
        print(f"{h / 1000:.5f}")
    return {"area_m2": area, "heights_m": heights, "descent": descent}

if __name__ == "__main__":
    results = main()
```